

Chapter 3

Energy and Energy Transfer

3.1 Thermodynamic concept of work

The work W done by, or on, a system is evaluated in terms of macroscopically observable forces and displacements

$$W = \int_{s_1}^{s_2} \mathbf{F} \cdot d\mathbf{s} \quad (3.1)$$

This is form of work called the **mechanical work**, and is used to estimate work done during compression and expansion of gas, stretching of liquid film etc. In thermodynamics, however, a broader interpretation of work is employed: *Work is done by a system on its surroundings if the sole effect of the energy transfer on everything external to the system could have been the raising of a weight.* In other words, if the energy transfer across system boundary is not heat, it must be work. An example

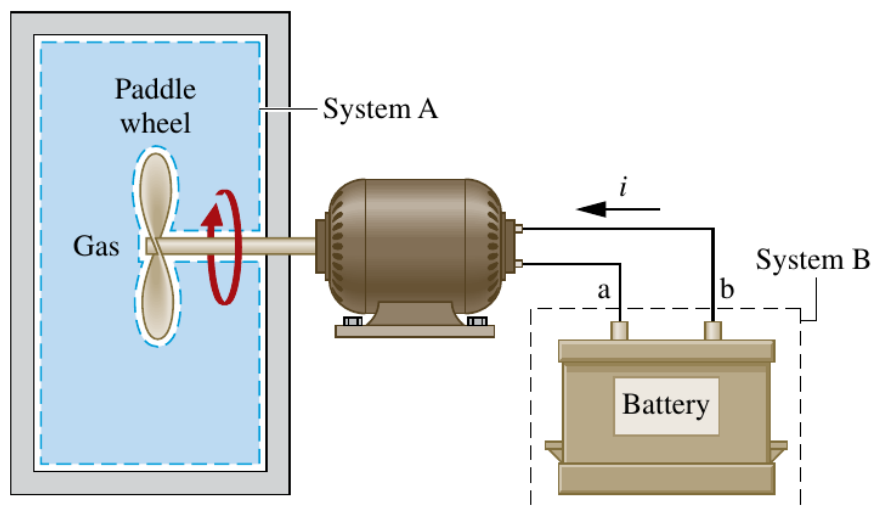


Figure 3.1: Example of electric work.

is illustrated in **Figure 3.1**. Here two systems, A and B, are shown. In system A gas is stirred by a paddle wheel doing mechanical work to transfer energy at the boundary between paddle wheel and the gas. In system B, however, there is no visible movement. Electricity moves across the boundary of system B. This electric current *could have raised a weight without any side effect*, and is therefore considered form of work in a broader sense of thermodynamics.

Heat transferred across a boundary also has the potential of *raising a weight* via a heat engine, but that will not be the sole effect. Heat engine will also reject heat to the surrounding as a side effect. Hence energy in the form of heat cannot be considered to be work.

Engineering thermodynamics follows the following sign convention for work:

$$W > 0 : \text{work done by the system} \quad (3.2)$$

$$W < 0 : \text{work done on the system} \quad (3.3)$$

Consider a system of expanding air-fuel mixture inside an internal combustion engine. Here work is done by the system on the surroundings so $W > 0$. If the air-fuel mixture inside the cylinder compresses (reduction in volume), the work is done by the surrounding on the system, hence $W < 0$.

To evaluate the integral given in Equation (3.1), it is necessary to know how the force varies with the displacement. This brings out an important idea about work: The value of W depends on the details of the interaction taking place between the system and the surroundings during a process and not just initial and final states of the system. It naturally then follows that **work is not a property** of the system or the surroundings. The differential work, δW is said to be *inexact* because, in general, the following integral is invalid:

$$\int_1^2 \delta W = W_2 - W_1 \quad (3.4)$$

On the other hand, the differential of a property is said to be *exact* because the change in a property between two particular states depends in no way on the details of the process linking the two states. For example, the change in volume between two states can be determined by integrating the differential dV , without regard for the details of the process, as follows

$$\int_1^2 dV = V_2 - V_1 \quad (3.5)$$

Here, V_1 is the volume at state 1 and V_2 is the volume at state 2. The differential of every property is exact. Exact differentials are written using the symbol d . To stress the difference between exact and inexact differentials, the differential of work is written as δW . The symbol δ is also used to identify other inexact differentials that will be encountered later.

3.2 Modeling expansion or compression work

There are many ways in which a system can do work. We start with an example of work done when a volume of a quantity of a gas changes by expansion or compression. Lets calculate the work done by a closed system shown in Figure 3.2, which consists of a gas contained in a piston-cylinder assembly. As the gas expands, it exerts a normal force on the piston which is simply equal to pA , where A is the area of the piston face. The work done by the system in displacing the piston by a distance dx is

$$\delta W = pA dx \quad (3.6)$$

The product $A dx$ in Equation (3.6) equals the change in volume of the system dV . Thus the work expression can be written as:

$$\delta W = p dV \quad (3.7)$$

Since dV is positive when volume increases, the work at the moving boundary is positive when the gas expands. During compression, the work becomes negative. For a change in volume from V_1 to

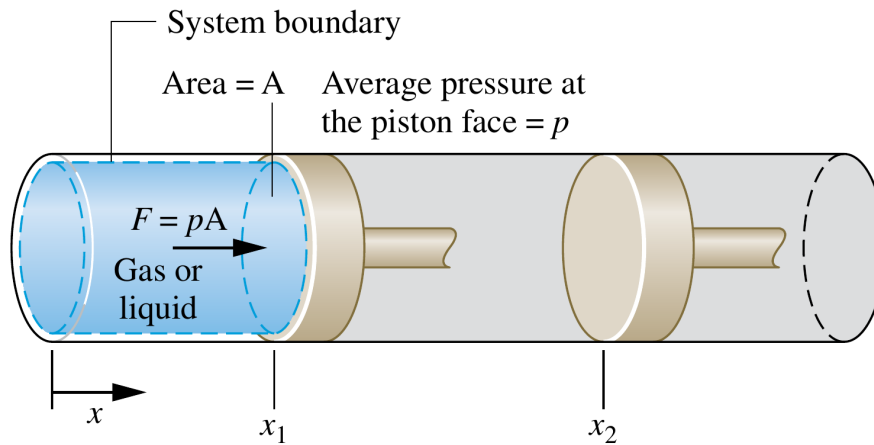


Figure 3.2: Schematic for calculating work due to expansion or compression of a gas.

V_2 , the work is obtained by integrating Equation (3.7):

$$W = \int_{V_1}^{V_2} p dV \quad (3.8)$$

Although Equation (3.8) is derived for the case of a gas in piston-cylinder assembly, it is applicable to systems of *any* shape provided the **pressure is uniform** with position over the moving boundary.

This condition of uniformity is present in quasistatic processes. In **quasistatic (or quasiequilibrium)** processes, intensive properties, such as pressure, are uniform spatially (with position) at any given instant of time. However, they vary temporally (with time).

Real-world processes are seldom quasistatic. In non-quasistatic processes, the intensive properties vary with position (spatial variation) at a given time. Also at a specified position, intensive properties may vary with time, sometimes chaotically. To perform the integral given in Equation (3.8), we require a relationship between gas pressure at the moving boundary and the system volume. However, due to non-quasiequilibrium effects during an *actual* expansion or compression process, this relationship may be difficult or almost impossible to obtain. An inherent assumption in that integral is that pressure is uniform throughout the volume of the system, especially when the system boundary is of arbitrary shape. However, in actual processes there may be nonuniformities in pressure.

Quasiequilibrium or quasistatic processes are idealization of real processes. A quasiequilibrium process is the one in which the departure from thermodynamic equilibrium is at most infinitesimal. All states through which the system passes in a quasiequilibrium process may be considered equilibrium states. Because nonequilibrium effects are inevitably present during actual processes, systems of engineering interests can at best approach, but never realize, a quasiequilibrium process. As an example consider again the system shown in Figure 3.2. During a quasiequilibrium process, which proceeds extremely slowly, system volume change very slightly at each step. The gas inside the cylinder gets sufficient time to adjust and attain a new equilibrium state, where all intensive properties, including pressure, are uniform spatially. Thus the relation given in Equation (3.8) is applicable to quasiequilibrium processes. Consider Figure 3.3(a) where the piston face is at position x_1 , and gas pressure is p_1 ; at the conclusion of a quasiequilibrium expansion process the piston face is at position x_2 , and the pressure is reduced to p_2 . At *each* intervening piston position, the uniform pressure throughout the gas is shown as a point on the diagram. The curve, or *path*, connecting states 1 and 2 on the diagram represents the equilibrium states through which the system has passed during the process. The work done by the gas on the piston during expansion is given by $\int p dV$, which can be interpreted as the area under the curve of pressure versus volume.

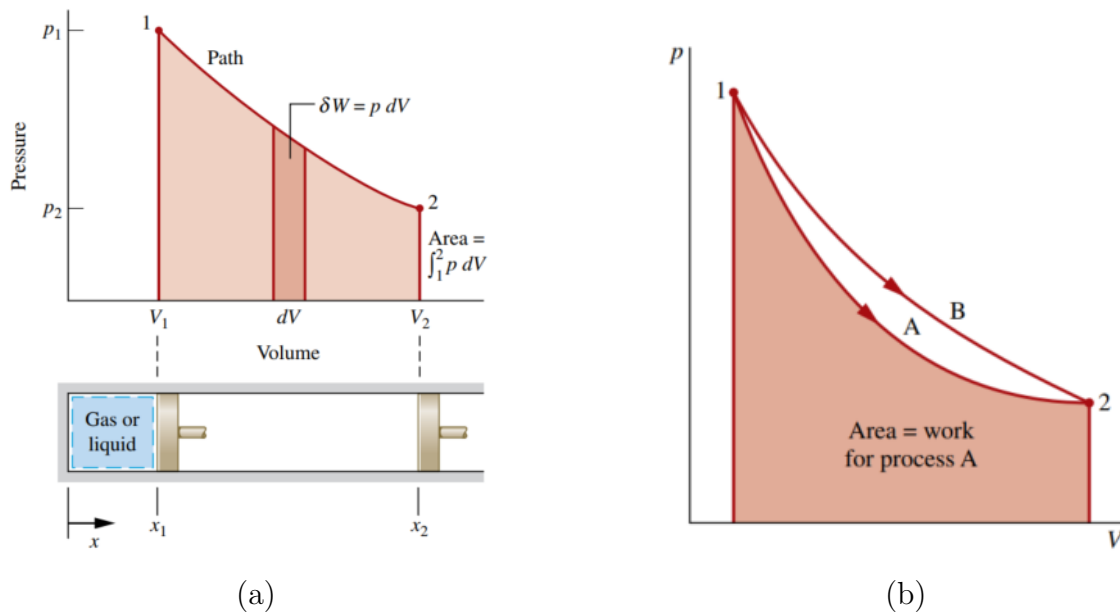


Figure 3.3: (a) Work of a quasiequilibrium expansion or compression process. (b) Illustration that work depends on the process.

The area interpretation of work in a quasiequilibrium expansion process allows a simple demonstration of the idea that **work depends on the process**. This can be brought out by referring to Figure 3.3(b). Suppose the gas in a piston-cylinder assembly goes from an initial equilibrium state 1 to a final equilibrium state 2 along two different paths, labeled A and B on Figure 3.3(b). Since the area beneath each path represents the work for that process, the work depends on the details of the process as defined by the particular curve and not just on the end states.

3.3 Thermodynamics concept of energy

In engineering thermodynamics, the change in the **total energy** of a system is considered to be made up of three *macroscopic* contributions. One is the change in kinetic energy, associated with the motion of the system *as a whole* relative to an external coordinate frame. Another is the change in gravitational potential energy, associated with the position of the system *as a whole* in Earth's gravitational field. All other energy changes are lumped together in the **internal energy** of the system. Like kinetic energy and gravitational potential energy, *internal energy is an extensive property* of the system, as is the total energy.

Internal energy is represented by the symbol U , and the change in internal energy in a process is $U_2 - U_1$. The specific internal energy is symbolized by u or $\bar{u}$ respectively, depending on whether it is expressed on a unit mass or per mole basis. the change in total energy is

$$E_2 - E_1 = (U_2 - U_1) + (KE_2 - KE_1) + (PE_2 - PE_1) \quad (3.9)$$

or,

$$\Delta E = \Delta U + \Delta KE + \Delta PE \quad (3.10)$$

To further our understanding of internal energy, consider a system consisting of a gas contained a tank. Let us develop a **microscopic interpretation of internal energy** by thinking of the energy attributed to the motions and configurations of the individual molecules, atoms, and subatomic particles making

up the matter of the system. Gas molecules move about, encountering other molecules or the walls of the container. Part of the internal energy of the gas is the *translational* kinetic energy of the molecules. Other contributions to the internal energy include the kinetic energy due to *rotation* of the molecules relative to their centers of mass and the kinetic energy associated with *vibrational* motions within the molecules. In addition, energy is stored in the chemical bonds between the atoms that make up the molecules. Energy storage on the atomic level includes energy associated with electron orbital states, nuclear spin, and binding forces in the nucleus. In dense gases, liquids, and solids, intermolecular forces play an important role in affecting the internal energy.

3.4 Energy Transfer by Heat

Thus far we have only seen energy transfer in the form of work between surroundings and *closed systems*. However, energy can also be transferred between them in the form of **heat**. Energy transfer due to heat occurs when there is a temperature difference between system and surroundings. The direction of heat transfer is in the direction of decreasing temperature.

The symbol Q denotes an amount of energy transferred across the boundary of a system in a heat interaction with the system's surroundings. Heat transfer *into* a system is taken to *positive*, and heat transfer *from* a system is taken as *negative*. When $Q > 0$, energy is transferred into the system. This is in contrast to the sign convention followed for work, where $W > 0$ denotes the energy transfer from the system to its surroundings. The value of heat transfer depends on the details of a process and not just the end states. Thus, like work, **heat is not a property**, and its differential is written as δQ . The amount of energy transfer by heat for a process is given by the integral

$$Q = \int_1^2 \delta Q \quad (3.11)$$

where the limits mean “from state 1 to state 2” and do not refer to the values of heat at those states. Just as for work, the notion of “heat” at a state has no meaning, and the integral should *never* be evaluated as $Q_2 - Q_1$.

The word **adiabatic** means *without heat transfer*. Thus if a system undergoes a process involving no heat transfer with its surroundings, that process is called an *adiabatic process*.

There are two basic methods of heat transfer: **conduction** and **radiation**. In addition, relationships are also available for evaluating energy transfer involving a *combined* mode called **convection**. Energy transfer by *conduction* can take place in solids, liquids and gases, or between any two phases if they are in contact. Conduction can be thought of as the energy transfer from more energetic particles of a substance to adjacent particles that are less energetic due to interaction between particles. By **Fourier's Law** the rate of heat transfer along x direction, denoted by $\dot{Q}_x$, is proportional to wall area A , and the temperature gradient in the x direction, dT/dx :

$$\dot{Q}_x = -\kappa A \frac{dT}{dx} \quad (3.12)$$

where the proportionality constant, κ is called the **thermal conductivity**. The minus sign is a consequence of energy transfer in the direction of *decreasing* temperature. Substances with large values of thermal conductivity such as copper are good conductors, and those with small conductivities are good insulators.

Thermal radiation is emitted by matter as a result of changes in electronic configurations of the atoms or molecules within it. The energy is transported by electromagnetic waves (or photons).

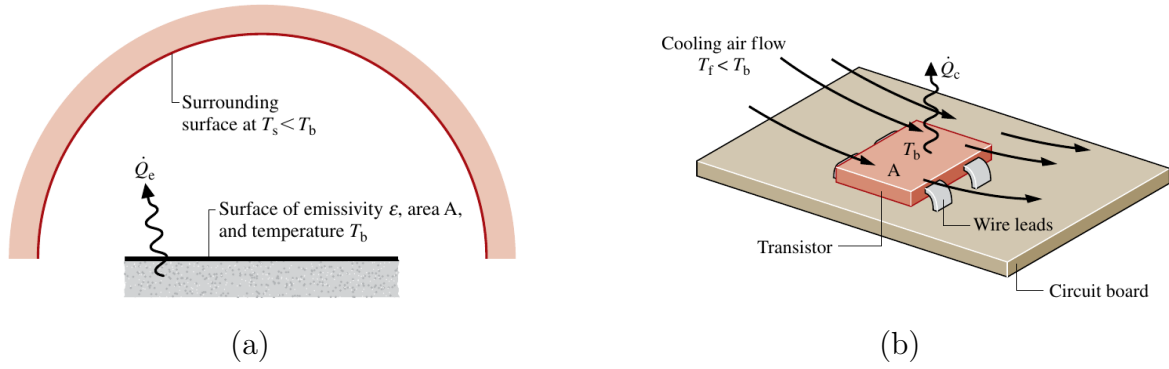


Figure 3.4: (a) Net radiation exchange (b) Heat exchange due to convection.

Unlike conduction, thermal radiation requires no intervening medium to propagate and can even take place in a vacuum. Solid surfaces, gases, and liquids all emit, absorb, and transmit thermal radiation to varying degrees. The rate at which radiation energy is emitted, $\dot{Q}_e$, from a surface of area A is quantified macroscopically by a modified form of the **Stefan-Boltzmann law**.

$$\dot{Q}_e = \epsilon \sigma A T_b^4 \quad (3.13)$$

which shows that the thermal radiation is associated with the fourth power of the absolute temperature of the surface, T_b . The emissivity, ϵ , is a property of the surface that indicates how effectively the surface radiates ($0 \leq \epsilon \leq 1.0$), and σ is the Stefan-Boltzmann constant:

$$\sigma = 5.67 \times 10^{-8} \text{W/m}^2 \cdot \text{K}^4 \quad (3.14)$$

In general the *net* rate of energy transfer by thermal radiation between two surfaces involves relationships among the properties of the surfaces, their orientations with respect to each other, the extent to which the intervening medium scatters, emits, and absorbs thermal radiation, and other factors. For example for a simple case of radiation exchange between a surface at temperature T_b and a much larger surrounding surface at T_s , the steady-state value of net rate of radiant exchange between the smaller surface, whose area is A and emissivity is ϵ , and the larger surrounding is

$$\dot{Q}_e = \epsilon \sigma A [T_b^4 - T_s^4] \quad (3.15)$$

The example is illustrated in **Figure 3.4(a)**.

Energy transfer between a solid surface at a temperature T_b and an adjacent gas or liquid at another temperature T_f and in motion plays a prominent role in the performance of many devices of practical interest. This is commonly referred to as **convection**. An example is illustrated in **Figure 3.4(b)**, where $T_b > T_f$, where energy is transferred in the direction indicated by the arrow and is due to the *combined* effect of conduction from solid to air and bulk motion of the air. The rate of energy transfer from the surface to the air can be quantified by the following *empirical* expression:

$$\dot{Q}_c = hA(T_b - T_f) \quad (3.16)$$

which is known as **Newton's law of cooling**. In **Equation (3.16)**, A is the surface area and the proportionality factor h is called the *heat transfer coefficient*. The heat transfer coefficient is *not* a thermodynamic property. It is an empirical parameter that incorporates into the heat transfer relationship the nature of the flow pattern near the surface, the fluid properties, and the geometry. When fans or pumps cause the fluid to move, the value of the heat transfer coefficient is generally greater than when relatively slow buoyancy-induced motions occur. These two categories are called *forced* and *free* (or *natural*) convection, respectively.

3.5 First law of thermodynamics

The only ways the *total energy* of a **closed system** can be changed are through transfer of energy by work or by heat. Further, based on the experiments of Mayer, Joule and others, a fundamental aspect of energy concept is that energy is conserved, which we call as the **first law of thermodynamics**.

$$\left[\begin{array}{c} \text{change in the amount} \\ \text{of total energy contained} \\ \text{within a system during} \\ \text{some time interval} \end{array} \right] = \left[\begin{array}{c} \text{net amount of energy} \\ \text{transferred in across} \\ \text{the system boundary by} \\ \text{heat transfer during} \\ \text{the time interval} \end{array} \right] - \left[\begin{array}{c} \text{net amount of energy} \\ \text{transferred out across} \\ \text{the system boundary by} \\ \text{work transfer during} \\ \text{the time interval} \end{array} \right]$$

The **total energy balance** for a *closed system* can be expressed in symbols as

$$E_2 - E_1 = Q - W \quad (3.17)$$

Introducing **Equation (3.10)**, we obtain an alternative form of the total energy balance equation:

$$\Delta KE + \Delta PE + \Delta U = Q - W \quad (3.18)$$

Various special forms of energy balance can be written. The energy balance in differential form is:

$$dE = \delta Q - \delta W \quad (3.19)$$

where dE is the differential form of *total energy*, a property. Since Q and W are not properties, their differentials are written as δQ and δW , respectively. The instantaneous **time rate form** of total energy balance is:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} \quad (3.20)$$

This can be further expanded as:

$$\frac{dKE}{dt} + \frac{dPE}{dt} + \frac{dU}{dt} = \dot{Q} - \dot{W} \quad (3.21)$$

Equations (3.17), (3.18) and (3.21) are alternate total energy balance relations that most suitable for **closed systems**. Although, these relations are applicable to control volumes (open systems) as well, better forms of these relations are available that can be applied to control volumes. In problems that will be considered in this course, the changes in kinetic energy and potential energy are small and can be neglected with respect to changes in other quantities. Thus, unless stated explicitly in the problems, changes in kinetic energy and potential energy should be neglected.

3.5.1 Example Problem 01

Mass m of a certain gas is contained inside a piston-cylinder assembly. The gas undergoes a process for which the pressure-volume relationship is:

$$p V^n = \text{constant} \quad (3.22)$$

where $n > 1$. The initial pressure is p_1 , the initial volume is V_1 , and the final volume is V_2 . The change in specific internal energy of the gas in the process is Δu . Determine the net heat transfer during the process.

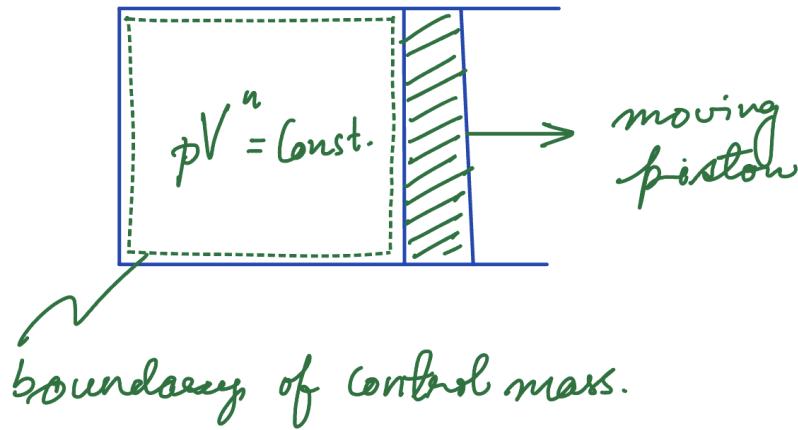


Figure 3.5: Schematic for example 01, which shows a control mass enclosed in dashed boundary.

Solution: The first step to the solution is to sketch the closed system (control mass) as shown in **Figure 3.5**. Since, no rates are involved, we will use **Equation (3.18)**:

$$\cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U = Q - W \quad (3.23)$$

The kinetic and potential energy changes drop out because they are usually small in comparison to other terms in the equation. The equation further becomes:

$$\begin{aligned} \Delta U &= Q - W \\ \Rightarrow m(\Delta u) &= Q - \int_{V_1}^{V_2} p dV \end{aligned} \quad (3.24)$$

The integral term in **Equation (3.24)** can be evaluated as follows:

$$\begin{aligned} \int_{V_1}^{V_2} p dV &= \int_{V_1}^{V_2} \frac{C}{V^n} dV \\ &= \frac{1}{1-n} \frac{C}{V^{n-1}} \Big|_{V_1}^{V_2} \\ &= \frac{1}{n-1} p V \Big|_{V_2}^{V_1} \\ &= \frac{1}{n-1} (p_1 V_1 - p_2 V_2) \end{aligned} \quad (3.25)$$

Since $p_2 = p_1 \left(\frac{V_1}{V_2}\right)^n$, the work done becomes:

$$\begin{aligned} \int_{V_1}^{V_2} p dV &= \frac{1}{n-1} \left(p_1 V_1 - p_1 \left(\frac{V_1}{V_2}\right)^n V_2 \right) \\ &= \frac{p_1 V_1}{n-1} \left(1 - \left(\frac{V_1}{V_2}\right)^{n-1} \right) \end{aligned} \quad (3.26)$$

Substituting **Equation (3.26)** into **Equation (3.24)**, we obtain:

$$Q = m(\Delta u) + \frac{p_1 V_1}{n-1} \left(1 - \left(\frac{V_1}{V_2}\right)^{n-1} \right) \quad (3.27)$$

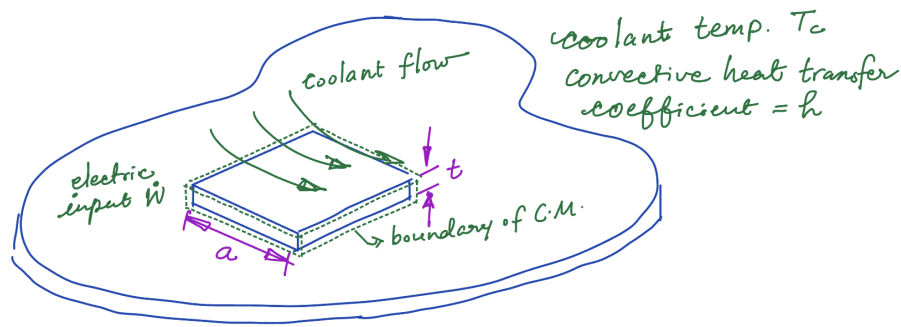


Figure 3.6: Schematic for example 02, where the control mass boundary is shown in dashed lines. The boundaries coincide with the outer surface of the chip.

Depending on the relative magnitudes of $\dot{V}_1$ and $\dot{V}_2$, and the value of Δu , the heat transferred Q can be positive or negative. If Q is positive, heat is supplied to the system, if it is negative, heat is lost to the surroundings.

3.5.2 Example Problem 02

A silicon chip can be modeled as a cuboid with a square top surface of side a and thickness t . At steady state, the chip has an electric power input of $\dot{P}$. The top surface of the chip is exposed to a coolant whose temperature is T_c . The heat transfer coefficient for convection between the chip and the coolant is h . If the heat transfer by conduction between the chip and the substrate is negligible, determine the surface temperature of the chip.

Solution: We first sketch the control mass of the system, whose boundaries coincide with the outer surface of the chip as shown in Figure 3.6. The chip is the control mass. Only energy can flow across its boundaries, not mass. Since power is involved in the problem, we use the time rate of change form of the first law given by Equation (3.20):

$$\frac{dE}{dt} = \dot{Q} - \dot{W} \quad (3.28)$$

The system is at steady-state so time rate of change of all properties will become zero, $dE/dt = 0$ in the present problem. The heat is lost from the system due to convection, thus the quantity $\dot{Q}$ is given by:

$$\dot{Q} = -h A (T_b - T_c) \quad (3.29)$$

where T_b is the temperature of the surface of the chip exposed to coolant. The energy balance, Equation (3.28), becomes:

$$0 = -h a^2 (T_b - T_c) - \dot{W} \quad (3.30)$$

Here, $\dot{W} = -\dot{P}$ since electric work is being done on the system and therefore,

$$T_b = T_c + \frac{\dot{P}}{h a^2} \quad (3.31)$$

Make a note of the sign of work and heat transfer rates. In the final solution, T_b has a value higher than that of T_c , which make sense, as the chip is supposed to have higher temperature than the coolant. The students are advised to use similar sanity checks after algebraically arriving at the final solution.

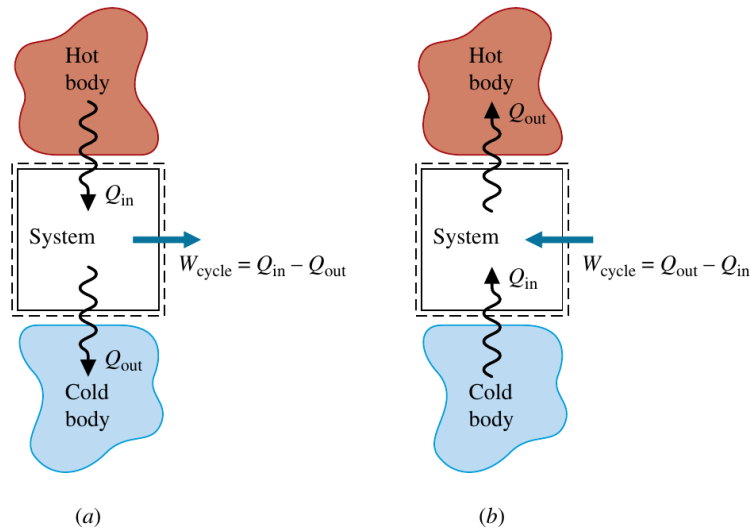


Figure 3.7: Schematic for applications of cycles: (a) Heat Engine/Power Cycle, (b) Refrigeration & Heat Pumps.

3.6 Thermodynamic Cycles

A **thermodynamic cycle** is a sequence of processes that begins and ends at the same state. At the conclusion of a cycle all properties have the same value they had at the beginning. Consequently, over the cycle the system experiences no *net* change of state. There are prominent roles of cycles in many areas of engineering applications such as power generation, vehicle propulsion, and refrigeration. Use of cycles as basic principles behind working of engineering applications make the equipment compact as well as practical. Devices based on cyclic principles can run for long durations of time and occupy a small amount of space, both of which are desirable from a user's perspective.

The *total energy balance* for a closed system undergoing a thermodynamic cycle takes the form.

$$\Delta E_{cycle} = Q_{cycle} - W_{cycle} \quad (3.32)$$

where Q_{cycle} and W_{cycle} represent *net* amounts of energy transfer by heat and work respectively, for the cycle. Since the system returned to its initial state after the cycle, there is no *net* change in its total energy, since it is a property. Therefore, the left side of **Equation (3.32)** equals zero, and the equation reduced to:

$$W_{cycle} = Q_{cycle} \quad (3.33)$$

Equation (3.33) is an expression of the conservation of energy principle that must be satisfied by *every* thermodynamic cycle. **Figure 3.7** are simplified schematics of two general classes of cycles: power cycles, also called heat engines, and refrigeration & heat pumps. In each case, a system undergoes a cycle while communicating thermally with two bodies – one hot and the other cold. These bodies are located in the surroundings of the system. During each cycle there is also a net amount of energy exchanged with the surroundings by work.

In power cycles, a net work is transferred to the surroundings in each cycle. Any such cycle is called a **power cycle** or a **heat engine**. The net work output equals the net heat input during the cycle:

$$W_{cycle} = Q_{in} - Q_{out} \quad (3.34)$$

Since in power cycle (heat engine), net work is done by the system, i.e., W_{cycle} is positive, Q_{in} must be greater than Q_{out} . In real world examples of power cycles, heat Q_{in} is supplied to the system by

burning fuel, or from a nuclear reaction; it can also be obtained from solar radiation. The heat Q_{out} is generally discharged to the surrounding atmosphere or a nearby body of water. The performance of a power cycle can be described in terms of the extent to which heat added Q_{in} is converted to net work output, W_{cycle} . The extent of conversions is represented by the following ratio called **thermal efficiency** :

$$\eta = \frac{W_{cycle}}{Q_{in}} = \frac{Q_{in} - Q_{out}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}} \quad (3.35)$$

We will see later that due to second law of thermodynamics the entire Q_{in} cannot be converted to work W_{cycle} , which ensures that η stays noticeably below unity.

Next, we consider the **refrigeration** and **heat pump cycles** shown in **Figure 3.7(b)**. The purpose of devices based on these cycles is to extract heat Q_{in} from the cold body and discharge heat Q_{out} to hot body. Both hot body and cold body belong to the surrounding of the system. To accomplish this energy transfer, work input W_{input} is required:

$$\begin{aligned} W_{cycle} &= Q_{in} - Q_{out} \\ -W_{input} &= Q_{in} - Q_{out} \\ W_{input} &= Q_{out} - Q_{in} \end{aligned} \quad (3.36)$$

Since, work (electric energy) is consumed by refrigeration devices, W_{input} is positive, and thus Q_{out} should be greater than Q_{in} . Although, the refrigerators and heat pumps are represented using the same schematic shown in **Figure 3.7(b)**, the objectives of a refrigerator and a heat pump are quite different. The objective of a **refrigeration cycle** is to cool a refrigerated space or to maintain the temperature within a dwelling or other building *below* that of the surroundings. The objective of a **heat pump** is to maintain the temperature within a dwelling or other building *above* that of the surroundings or to provide heating for certain industrial processes that occur at elevated temperatures.

Since refrigeration and heat pump cycles have different objectives, their performance parameters, called **coefficients of performance (COP)**, are defined differently. The performance of **refrigeration cycles** can be described as the ratio of the amount of energy received by the system undergoing the cycle from the cold body, Q_{in} , to the net work into the system to accomplish this effect, W_{input} . Thus the coefficient of performance, β , is

$$\beta = \frac{Q_{in}}{W_{input}} = \frac{Q_{in}}{Q_{out} - Q_{in}} \quad (3.37)$$

For a household refrigerator, Q_{out} is discharged to the space around the refrigerator. Work W_{input} is usually provided in the form of electricity to run the motor that drives the refrigerator.

The performance of **heat pumps** can be described as the ratio of the amount of energy discharged from the system undergoing the cycle to the hot body, Q_{out} , to the net work into the system to accomplish this effect, W_{input} . Thus, the coefficient of performance, γ , is

$$\gamma = \frac{Q_{out}}{W_{input}} = \frac{Q_{out}}{Q_{out} - Q_{in}} \quad (3.38)$$

From this equation it is seen that the COP γ is never less than unity. For residential heat pumps, the energy quantity Q_{in} is normally drawn from the surrounding atmosphere, the ground, or a nearby body of water. Work W_{input} is usually provided by electricity.

The coefficient of performances β and γ are defined as ratios of the desired heat transfer effect to the cost of work to accomplish that effect. It is therefore, desirable that these coefficients have values that are as large as possible.

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